

Post-Exploitation Enumeration Notes

Additional Concepts Learned — Linux Systems, Filesystems & Environment

1. Operating System vs Linux Kernel

At first these seemed like the same thing. They are not.

- **Operating System (Ubuntu)** — the complete operating system.
- **Linux Kernel** — the core component inside the operating system.

Example from the Lab

Component	Value
OS	Ubuntu 8.04
Kernel	Linux 2.6.24-16-server

Analogy: Car = Operating System | Engine = Kernel

The Kernel Manages

- CPU
- Memory
- Processes
- Hardware
- Networking
- Filesystems

The Operating System Also Includes

- Bash
- Package manager (apt)
- Libraries
- Utilities
- Applications

2. Why Enumerate Both?

They answer different questions.

Operating System Tells Me

- Which Linux distribution I'm on
- Package manager (apt, yum, etc.)
- Typical file locations
- Overall software ecosystem

Kernel Tells Me

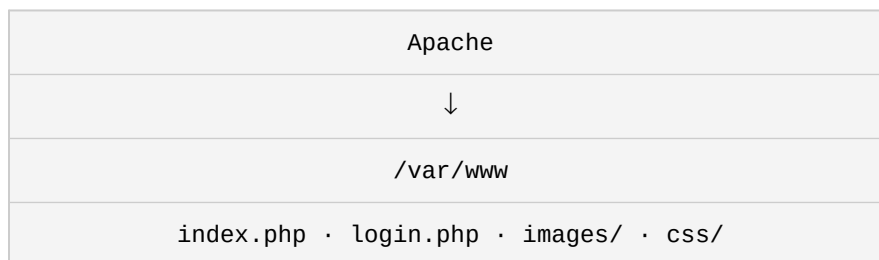
- Which operating system core is running
- Potential kernel vulnerabilities
- Kernel capabilities

General Enumeration Priority

- Service versions
- Kernel version
- Operating System

3. /var/www

/var/www is Apache's web root. It contains the actual website files served to users.



In Metasploitable, the following were physically stored inside **/var/www**:

- DVWA
- Mutillidae
- phpMyAdmin
- TikiWiki

4. /tmp vs /var/tmp

/tmp — Temporary Workspace

Programs use it while they are running.

- Temporary downloads
- Installer files
- Browser temporary files

- Program working files

Usually cleared after reboot.

/var/tmp

Also stores temporary files. **Difference:** files usually survive reboot. Used when temporary data needs to persist longer.

5. Why /tmp Exists

Purpose: provide programs with a temporary workspace. Instead of filling permanent directories with temporary files, applications store them in **/tmp**.

/home	/tmp
Cupboard (permanent)	Workbench (temporary)

6. Environment Variables

Environment variables are named values provided by the operating system that describe the current environment. Programs read them whenever they need information.

Examples

```
PATH
PWD
HOME
TERM
REMOTE_HOST
```

Not only the shell uses them — **any** program can read environment variables if needed:

- Bash
- Vim
- SSH
- Python
- Web servers
- Databases

7. PATH (Detailed Understanding)

Commands are simply executable files. For example, **ls** is actually **/bin/ls**.

When typing **ls**, Bash receives the string "ls". Since there is no slash (/), Bash checks **PATH**.

```
PATH=/usr/local/bin:/usr/bin:/bin
```

Bash searches in order:

<code>/usr/local/bin/ls</code>
↓
<code>/usr/bin/ls</code>
↓
<code>/bin/ls</code>
↓
Found

Then Bash asks the kernel to execute it.

Without PATH: the shell would have no idea where `ls` exists — I would have to type `/bin/ls` every single time.

PATH therefore exists so the shell knows where to search for programs.

8. Main Idea Behind Environment Variables

Initially I thought they mainly existed to locate resources. More accurate definition:

Environment variables provide programs with information about the environment they are running in.

Variable	Meaning
PATH	Where commands are located
PWD	Current directory
HOME	User's home directory
TERM	Terminal capabilities
REMOTE_HOST	Remote machine

PATH is only one example. The broader concept is information about the execution environment.

9. Linux Mounting

Mounting means attaching a storage device or filesystem to a directory.

Windows

C:
D:
E:

Linux

```
/
├── home
├── var
├── media
└── mnt

USB → /mnt/usb
```

Instead of receiving a new drive letter, storage becomes part of the existing filesystem.

10. Why Linux Uses Mounting

The major advantage: programs always use the same path.

Example: a website stores uploads at **/uploads**. If the original disk becomes full, the administrator mounts a new disk at **/uploads**. The application continues using **/uploads** without knowing the underlying storage changed.

Linux separates the **logical path** from the **physical storage**.

11. What Is a “New Disk”?

Usually another storage device:

- Internal SSD
- Internal HDD
- External hard drive
- USB drive
- Network storage (NFS)

Linux can mount any of these into the existing filesystem.

12. Why We Ran mount During Enumeration

We did **not** mount anything. The command **mount** lists currently mounted filesystems.

Purpose: understand storage layout. It tells us:

- Which storage devices exist
- Where they are mounted
- Filesystem type
- Mount options

Example Output

```
/dev/mapper/metasploitable-root on / type ext3
```

Storage device	/dev/mapper/metasploitable-root
Mounted at	/
Filesystem	ext3

13. ext3, ext4, NTFS, vfat

These are simply different filesystem types. Their job: organize and manage files on storage devices.

Filesystem	Description
ext3	Older Linux filesystem
ext4	Newer Linux filesystem
NTFS	Windows filesystem
vfat (FAT32)	Common on USB drives

Think of them as different methods of organizing files on a disk.

14. /etc/hosts

Purpose: local hostname → IP address mapping.

```
192.168.1.10 fileserver
```

When running **ping fileserver**, Linux first checks **/etc/hosts**. If found, it uses the IP immediately — no DNS request is made. Only if not found does Linux query the DNS server.

Important: it is NOT a DNS cache. It is a manually maintained lookup table.

Search Order

Program
↓
/etc/hosts
↓ (if not found)
DNS Server

15. Attacker Perspective

Interesting entries inside **/etc/hosts** can reveal internal systems.

```
10.0.0.5 database
10.0.0.6 backup
```

Even if DNS is unavailable, an attacker now knows those internal systems exist.

Key Takeaways

- Operating System and Kernel are different — the kernel is the core of the operating system.
- Service versions are usually a higher enumeration priority than OS version.
- **/var/www** stores website files.
- **/tmp** is temporary workspace; **/var/tmp** stores temporary files that usually survive reboot.
- Environment variables describe the current execution environment.
- PATH tells Bash where to search for executables.
- Environment variables are used by many different programs, not only Bash.
- Mounting attaches storage devices to directories.
- Linux integrates storage into one filesystem tree instead of using drive letters.
- **mount** is an enumeration command that reveals the storage layout.
- **ext3**, **ext4**, **NTFS**, and **vfat** are different filesystem types.
- **/etc/hosts** provides local hostname-to-IP mappings and is checked before DNS.